Use OpenNMT-py library to design a machine translation (MT) system. For more information, please check https://github.com/OpenNMT/OpenNMT-py

For building an MT system, you need parallel sentences or sentence pairs in source and target languages. The dataset is available at https://huggingface.co/datasets/ai4bharat/BPCC

Take **ilci** subset from this data. Choose your native language as the target language while English is the source language. Create train and test splits in the dataset. While designing the data splits, do not just randomly choose sentences. The selection of the sentences should be based on lengths and each split should consist of sentences of varying lengths. Use 4 bins while creating the datasets (both train and test).

bin1 - sentence length <= 10
bin2 - sentence length in (10, 20)
bin3 - sentence length in (20, 30)
bin4 - sentence length >= 30

Design an neural machine translation model on the training data and evaluate it on the test data. Use BLEU metric for translation evaluation. Use https://github.com/mjpost/sacrebleu for computing BLEU. Provide bin wise BLEU scores. Report your bin-wise result analysis.